

Contact Management System – Detailed Algorithm

1. **Start** the program.
2. **Initialize** an empty contact list (ArrayList or database).
3. Display the **Main Menu**:
 - Add Contact
 - View All Contacts
 - Search Contact
 - Update Contact
 - Delete Contact
 - Sort Contacts
 - Exit
4. Read the user's menu choice.
5. If the choice is **Add Contact**:
 - Enter Contact ID.
 - Enter Name.
 - Enter Phone Number.
 - Enter Email Address.
 - Enter Address.
 - Check whether the Contact ID or Phone Number already exists.
 - If a duplicate is found, display "**Contact Already Exists.**"
 - Otherwise, save the contact successfully.
6. If the choice is **View All Contacts**:
 - Check whether the contact list is empty.
 - If empty, display "**No Contacts Available.**"
 - Otherwise, display all contact details.
7. If the choice is **Search Contact**:
 - Enter the Contact ID, Name, or Phone Number.
 - Search the contact list.

- If found, display the contact details.
- Otherwise, display "**Contact Not Found.**"

8. If the choice is **Update Contact**:

- Enter the Contact ID or Phone Number.
- Search for the contact.
- If found, edit the required details.
- Save the updated information.
- Display "**Contact Updated Successfully.**"
- Otherwise, display "**Contact Not Found.**"

9. If the choice is **Delete Contact**:

- Enter the Contact ID or Phone Number.
- Search for the contact.
- If found, delete the contact.
- Display "**Contact Deleted Successfully.**"
- Otherwise, display "**Contact Not Found.**"

10. If the choice is **Sort Contacts**:

- Sort contacts alphabetically by name.
- Display the sorted contact list.

11. If the choice is **Exit**:

- Save data if required.
- Display "**Thank You for Using Contact Management System.**"

12. If an invalid option is entered:

- Display "**Invalid Choice. Please Try Again.**"

13. Return to the **Main Menu** after completing any operation.

14. Repeat the process until the user selects **Exit**.

15. **Stop** the program.